

Exp. No.: 7	Identify Singular Configurations of the Robot
Date:	

Aim:

To invoke options of enabling virtual fence around the robot to prevent it from passing a defined plane.

Procedure:

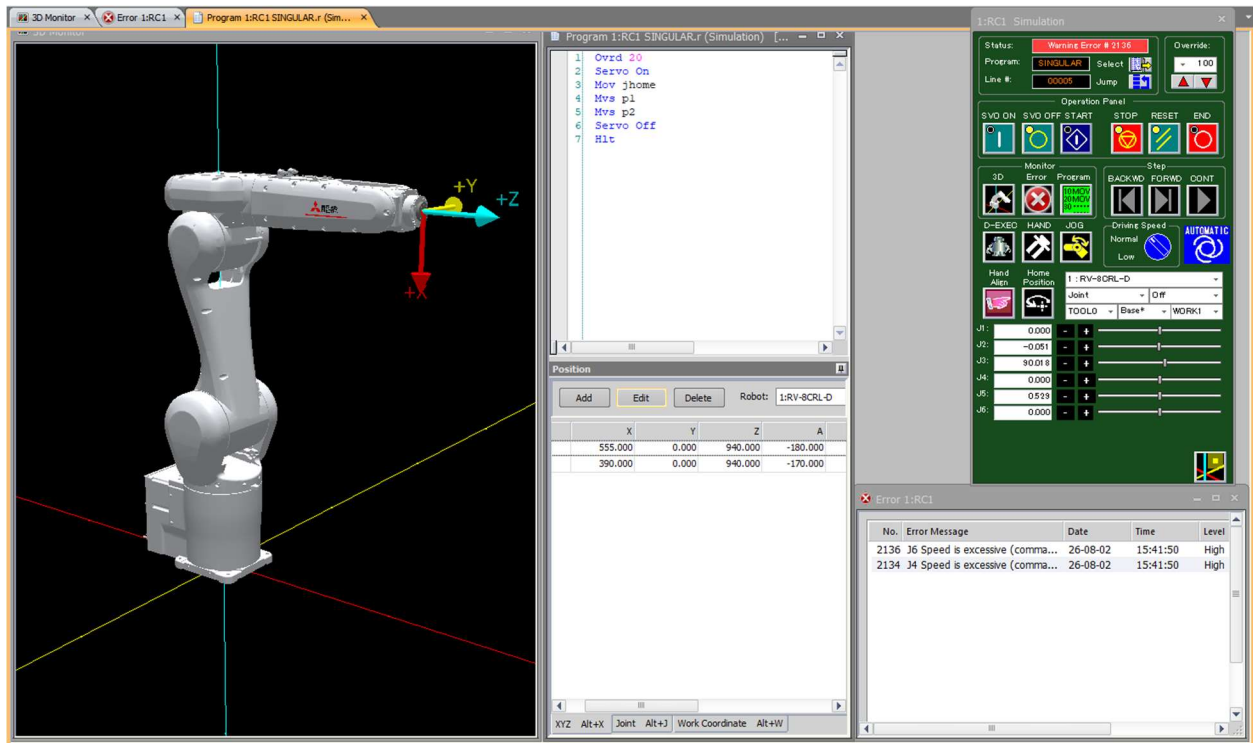
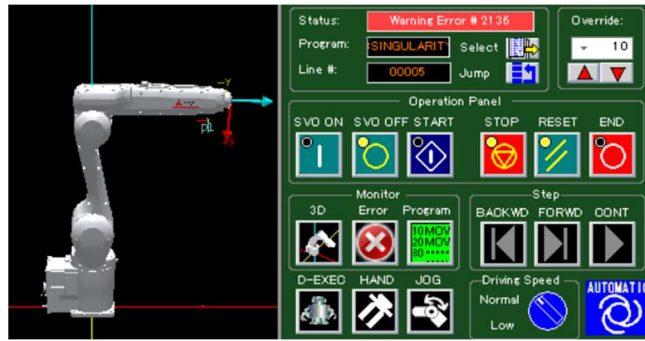
- Create a new project, > select the robot's name > Series CR800-D > 6 axis > 8 kg load > Model: RV-8CRL-D > Finish
- Select default settings while creating new workspace for this experiment.
- Select Simulator if Offline.
- Define a home position p_{home} at (470, 0, 855, -180, 0, 180) with no flags FLG1 or FLG2 or j_{home} at (0, 0, 90, 0, 90, 0) for the robot.
- Define two other points p₁ and p₂ or j₁ and j₂ such a way that when robot moves from configuration 1 to configuration 2, it shows a singularity error.
- Show that each point p₁ or j₁ and p₂ or j₂ are individually reachable within the workspace by the robot tool. However, the singularity occurs when moving from one point to the other.
- Write a program to travel from j_{home} to j₁ or p₁ and then to j₂ or p₂ that may show the error as J4 and J6 'Speed is excessive'.
- Send the robot back to its home position once the verification is finished.

RT Toolbox3 Program:

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Observation:

- The controller gives an error # 2136 and #2134 indicating 'J6 Speed is excessive' and 'J4 Speed is excessive' respectively when travelling from p₁ to p₂, which are representative of the singularities associated with the selected configurations.



Results & Inference: